

ALISON EMILIEN

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PROFESSIONAL SUMMARY

Software Developer with a BSc in Electrical Engineering and a BSc in Computing Technology. Skilled in Python, C/C++, JavaScript, testing, debugging, and Linux, with project experience in web applications, automation, AI systems, and embedded software. I have demonstrated teamwork and organisation skills through my work experiences.

SKILLS

Languages: Python, C, C++, JavaScript, C#, .NET, SQL

Web and APIs: Vite, Streamlit, HTML/CSS, REST API integration, JSON

Software Engineering: Object-oriented programming, data structures and algorithms, pytest, unit testing, debugging, event-driven design, SDLC

Tools and Platforms: Git, GitHub, Linux, Docker, Vercel CI/CD, ROS 2, OpenCV, LangChain, ChromaDB, OpenAI API, SSMS, VS Code, Rider

Spoken Languages: French, English and Mauritian Creole

PROJECTS

Engineering Portfolio Website: <https://github.com/Alison0704/portfolio-website>

- Developed a professional portfolio website showcasing technical projects, skills and engineering experiences.
- Implemented keyword-based project filtering, technology tags, and direct GitHub repository links.
- Established a CI/CD workflow using GitHub and Vercel for version control and automated builds.

ChipChat RAG Assistant: <https://github.com/Alison0704/ChipChat>

- Developed a Streamlit-based RAG assistant for ASIC design and verification questions.
- Built a semantic retrieval pipeline using PyPDFLoader, OpenAI embeddings, ChromaDB, and LangChain.
- Integrated GPT-4o-mini with hardware-focused prompting, caching, chat history, and exception handling.

Autonomous Hosting Robot: <https://github.com/Alison0704/Capstone-JimE>

- Developed a modular control application coordinating navigation, user interaction, diagnostics, and motor control.
- Implemented an event-driven finite-state machine and control loop for client, table, and return-home workflows.
- Integrated ROS2 publisher/subscriber interfaces, computer vision, sensors, a web interface, and an ESP32 controller.

WORK EXPERIENCE

Teaching Assistant - Digital Systems and C Programming

Jan 2024 - Apr 2026

University of Ottawa · Ottawa, ON

- Supported students in digital systems and digital circuit behavior, logic designs using Quartus II and Altera tools.
- Assisted students with debugging lab work and hardware behavior and connecting theory to practical digital designs.
- Helped students strengthen their problem-solving skills in logic design, C programming, and engineering debugging.

Pension and Benefits Assistant

May 2023 - Apr 2026

University of Ottawa · Ottawa, ON

- Used advanced Excel to build, maintain, and organize complex data logs with strong attention to accuracy.
- Handled sensitive pension and benefits information while following confidentiality and regulatory standards.
- Improved workflow visibility through structured documentation, tracking systems, and data quality checks.

EDUCATION

BSc Electrical Engineering - BSc Computing Technology

Sep 2020 – Apr 2026

University of Ottawa · Ottawa, ON